



Daffodil *International* **University**

ASSIGNMENT

Title: Final Assignment

Course Title: Numerical Methods

Course Code: CSE226

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1) Calculate the approximate value of $I = \int_0^6 (e^x + 2x) dx$ by using Trapezoidal, Simpson's $\frac{1}{3}$, Simpson's $\frac{3}{8}$ and Weddle's rule. Also Compute the Exact Value and Compare the results & Comment on it.

Solution:

Given, $I = \int_0^6 (e^x + 2x) dx$, let $y = f(x) = (e^x + 2x)$

Here, upper limit, $b = 6$,
lower limit, $a = 0$

Let, number of sub-intervals $n = 6$

Now, $h = \frac{6-0}{6} = 1$

The values of the function y at each sub-interval are given in the tabular form:

x	$x_0 = 0$	$x_1 = 1$	$x_2 = 2$	$x_3 = 3$	$x_4 = 4$	$x_5 = 5$	$x_6 = 6$
y	$y_0 = 1$	$y_1 = 4.7183$	$y_2 = 11.389$	$y_3 = 26.0855$	$y_4 = 62.5982$	$y_5 = 158.4132$	$y_6 = 415.4289$

Trapezoidal Rule

We know,

$$\int_{x_0}^{x_n} y dx = \frac{h}{2} [(y_0 + y_n) + 2(y_1 + y_2 + y_3 + \dots + y_{n-1})]$$

$$\therefore \int_0^6 (e^x + 2x) dx = \frac{1}{2} [(1 + 15.4288) + 2(4.7183 + 11.3891 + 26.0855 + 62.5982 + 158.4132)]$$
$$= 471.4187$$

Simpson's $\frac{1}{3}$ Rule

We know,

$$\int_0^6 (e^x + 2x) dx = \frac{h}{3} [(y_0 + y_6) + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4)]$$
$$= \frac{1}{3} [(1 + 15.4288) + 4(4.7183 + 26.0855 + 158.4132) + 2(11.3891 + 62.5982)]$$
$$= 440.4238$$

Simpson's $\frac{3}{8}$ Rule:

We know,

$$\int_0^6 (e^x + 2x) dx = \frac{3 \times h}{8} [(y_0 + y_6) + 3(y_1 + y_2 + y_4 + y_5) + 2(y_3)]$$

$$= \frac{3 \times 1}{8} [(1 + 415.4288) + 3(4.7183 + 11.3891 + 62.5982 + 158.4132) + 2(26.0855)]$$

$$= 442.4836$$

Weedle's Rule:

We know

$$\int_0^6 (e^x + 2x) dx = \frac{3h}{10} [(y_0 + y_2 + y_4 + y_6) + 5(y_1 + y_3 + y_5) + y_3]$$

$$= \frac{3 \times 1}{10} [(1 + 11.3891 + 62.5982 + 415.4288) + 5(4.7183 + 26.0855 + 158.4132) + 26.0855]$$

$$= 438.7760$$

The Exact value of $\int_0^6 x(e^x + 2x) dx$

$$\left[(e^x) \cdot \frac{x^2}{2} + (x + x^2) \cdot x \right]_0^6 = \left[\frac{e^x x^2}{2} + x^3 + x^4 \right]_0^6$$

$$= \left(\frac{e^6 \cdot 36}{2} + 216 + 1296 \right) - 0 = (e^6 \cdot 18 + 1512)$$

$$= 438.4288$$

Percentage error for Trapezoidal rule

$$= \left| \frac{438.4288 - 471.4187}{438.4288} \right| \times 100\%$$

$$= 7.4561\%$$

Percentage error for Simpson's $\frac{1}{3}$ Rule:

$$= \left| \frac{438.4288 - 440.4238}{438.4288} \right| \times 100\%$$

$$= 0.3866\%$$

Percentage error for Simpson's $\frac{3}{8}$ Rule:

$$= \left| \frac{438.4288 - 442.4836}{438.4288} \right| \times 100\%$$

$$= 0.8564\%$$

Percentage error for Weedle's Rule: $y_{\text{calc}} = Y$

$$= \left| \frac{438.4288 - 438.7760}{438.4288} \right| \times 100\%$$

$$= 0.0108\%$$

From the comparison above, it is noticable that Weedle's method gives the closest accurate result as compared to the exact value and the error percentage is just 0.0108%.

21 Estimate the value of $y(6)$ for which the following data fits the Exponential Function, $y = ae^{bx}$

x	1	2	3	4
y	3	5	7	10

Solution:

Given, exponential equation

$$y = ae^{bx} \quad \text{--- (1)}$$

Taking $\ln$ on the each side of eq (1) we get,

$$\ln y = \ln ae^{bx}$$

$$\Rightarrow \ln y = \ln a + \ln e^{bx}$$

$$\Rightarrow \ln y = \ln a + bx \ln e \quad \text{--- (2)}$$

Let, $Y = \ln y$

$A = \ln a$

$B = b \ln e$

So, the eq(2) can be written as

$$Y = A + Bx$$

The Normal Forms are-

$$\sum Y = nA + B \sum x \quad \text{--- (3)}$$

$$\sum xY = A \sum x + B \sum x^2 \quad \text{--- (4)}$$

Here, the number of data points of x , $n=4$

Calculation for finding the coefficient of a and b
we form the following table -

x	y	$Y = \ln y$	xY	x^2
1	3	1.0986	1.0986	1
2	5	1.6094	3.2188	4
3	7	1.9459	5.8377	9
4	10	2.3026	9.2104	16
$\sum x = 10$	$\sum y = 25$	$\sum Y = 6.9565$	$\sum xY = 19.3655$	$\sum x^2 = 30$

Putting these values in eq (3) and (4) we get

$$6.9565 = 4A + 10B \quad (5)$$

$$19.3655 = 10A + 30B \quad (6)$$

4 (6) $\times 2^1 - (5) \times 5^0$ we get

$$38.731 = 20A + 60B$$

$$34.7825 = 20A + 50B$$

$$3.9485 = 10B$$

$$B = 0.3949$$

Now,

$$B = b_{line} = 0.3949$$

$$\Rightarrow b = \frac{0.3949}{line}$$

Putting the value of B in (5)

$$6.9565 = 4A + 10(0.3949)$$

$$\Rightarrow 6.9565 = 4A + 3.949$$

$$\Rightarrow 4A = 3.0075$$

$$A = 0.7519$$

$$\therefore \ln a = 0.7519$$

$$\Rightarrow a = \text{ant} \ln(0.7519) \text{ or } e^{0.7519}$$

$$= 2.121$$

Putting these values of a, b in eq (1) we get

$$y = (2.121)e^{0.3949x}$$

for $y(6)$; $y = (2.121)e^{0.3949 \times 6}$

$$= 22.6756 \quad (\text{Ans})$$

31 Estimate the value of $y(5)$ for which the following data fits the Second Degree Polynomial using the following data:

x	1	2	3	4
y	2	4	6	9

Solution:

Let the equation of the Second Degree polynomial

$$y = a + bx + cx^2 \quad \text{--- (i)}$$

The normal equations are

$$\Sigma y = na + b \Sigma x + c \Sigma x^2 \quad \text{--- (ii)}$$

$$\Sigma xy = a \Sigma x + b \Sigma x^2 + c \Sigma x^3 \quad \text{--- (iii)}$$

$$\Sigma x^2 y = a \Sigma x^2 + b \Sigma x^3 + c \Sigma x^4 \quad \text{--- (iv)}$$

The number of data points, $n = 4$

To find the coefficient of a and b we form the below table:-

x	y	xy	x^2y	x^2	x^3	x^4
1	2	2	2	1	1	1
2	4	8	16	4	8	16
3	6	18	54	9	27	81
4	9	36	144	16	64	256
$\Sigma x = 10$	$\Sigma y = 21$	$\Sigma xy = 64$	$\Sigma x^2y = 216$	$\Sigma x^2 = 30$	$\Sigma x^3 = 100$	$\Sigma x^4 = 354$

Putting these values in eq (i), (ii) and (iii) we get $\Rightarrow$

$$21 = 4a + 10b + 30c \quad \text{--- (v)}$$

$$64 = 16a + 30b + 100c \quad \text{--- (vi)}$$

$$216 = 30a + 100b + 354c \quad \text{--- (vii)}$$

Now, (vi) $\times 2$ - (v) $\times 5 \Rightarrow$

$$128 = 20a + 60b + 200c$$

$$105 = 20a + 50b + 150c$$

$$\begin{array}{r} (v) \quad (v) \quad (v) \quad (v) \\ 128 \\ - 105 \\ \hline 23 = 10b - 50c \end{array} \quad \text{--- (viii)}$$

(vii) - (vi) $\times 3$

$$216 = 30a + 100b + 354c$$

$$192 = 30a + 90b + 300c$$

$$\begin{array}{r} (v) \quad (v) \quad (v) \quad (v) \\ 216 \\ - 192 \\ \hline 24 = 10b + 54c \end{array} \quad \text{--- (ix)}$$

$$(1x)^{10} = (1000)$$

$$24 = 10b + 54c$$

$$23 = 10b' - 50c$$

$$\begin{array}{r} (-) \quad (-) \quad (+) \\ \hline \end{array}$$

$$18 = 4c$$

$$c = \frac{1}{4} = 0.25$$

From (1x) $\Rightarrow$

From (1x) $\Rightarrow$

$$24 = 10b + 54 \times \frac{1}{4}$$

$$21 = 4a + 10 \times (1.05) + 30(0.25)$$

$$\Rightarrow 10b = 24 - \frac{27}{2} = \frac{21}{2}$$

$$\Rightarrow a = 0.75$$

$$\Rightarrow b = \frac{21}{2 \times 10} = 1.05$$

Putting all these values of a, b, c in eq (i) we get

$$y = 0.75 + 1.05x + 0.25x^2$$

For $y(5)$ or $x=5$

$$y = 0.75 + 1.05 \times (5) + 0.25 \times (5)^2$$

$$= 12.25$$

(Ans)

41 Use Runge-Kutta 4th order method to solve the ODE

$$10 \frac{dy}{dx} = x^2 + y^2, \quad y(0) = 1 \text{ for the interval } 0 \leq x \leq 2.$$

Solution:

Given ODE,

$$10 \frac{dy}{dx} = x^2 + y^2$$

$$\Rightarrow f(x, y) = \frac{dy}{dx} = \frac{x^2 + y^2}{10}$$

$$\text{Here, } x_0 = 0, \quad x_1 = 0.5, \quad (x_2 = 1.0, \quad x_3 = 1.5) \\ y_0 = 1$$

$$\text{Taking, steps } h = 0.5, \quad x_1 = 0.5$$

By the Runge-Kutta 4th order method,

$$y_{n+1} = y_n + \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4] \quad \text{--- ①}$$

where

$$k_1 = hf(x_n, y_n)$$

$$k_2 = hf\left(x_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right)$$

$$k_3 = hf\left(x_n + \frac{h}{2}, y_n + \frac{k_2}{2}\right)$$

$$k_4 = hf(x_n + h, y_n + k_3)$$

Step-1 putting $n=0$

$$k_1 = hf(x_0, y_0) \quad \dots \quad k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right)$$

$$= 0.5 \left[\frac{0^2 + 1^2}{10} \right]$$

$$= 0.05$$

$$= 0.5 f\left[0 + \frac{0.5}{2}, 1 + \frac{0.05}{2}\right]$$

$$= 0.5 f[0.25, 1.025]$$

$$= 0.5 \left[\frac{(0.25)^2 + (1.025)^2}{10} \right]$$

$$= 0.0557$$

$$k_3 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right)$$

$$= 0.5 f\left[0 + \frac{0.5}{2}, 1 + \frac{0.0557}{2}\right]$$

$$= 0.5 f(0.25, 1.0279)$$

$$= 0.5 \left[\frac{(0.25)^2 + (1.0279)^2}{10} \right]$$

$$= 0.0559$$

$$k_4 = hf(x_0 + h, y_0 + k_3)$$

$$= 0.5 f[0 + 0.5, 1 + 0.0559]$$

$$= 0.5 f(0.5, 1.0559)$$

$$= 0.5 \left[\frac{(0.5)^2 + (1.0559)^2}{10} \right] = 0.0682$$

$$\begin{aligned}\therefore Y_1 &= Y(0.5) = Y_0 + \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4] \\ &= 1 + \frac{1}{6} (0.05 + 2 \times 0.0557 + 2 \times 0.0559 + 0.0682) \\ &= 1.0569\end{aligned}$$

Step 2: Putting $n=1$

$$k_1 = hf(x_1, y_1)$$

$$= 0.5 f(0.5, 1.0569)$$

$$= 0.5 \left[\frac{(0.5)^2 + (1.0569)^2}{10} \right]$$

$$= 0.0683$$

$$k_2 = hf\left(x_1 + \frac{h}{2}, y_1 + \frac{k_1}{2}\right)$$

$$= hf\left(0.5 + \frac{0.5}{2}, 1.0569 + \frac{0.0683}{2}\right)$$

$$= hf(0.75, 1.09165)$$

$$= 0.5 \left[\frac{(0.75)^2 + (1.09165)^2}{10} \right]$$

$$= 0.0876$$

$$k_3 = hf\left(x_1 + \frac{h}{2}, y_1 + \frac{k_2}{2}\right)$$

$$= hf\left[0.5 + \frac{0.5}{2}, 1.0569 + \frac{0.0876}{2}\right]$$

$$= 0.5 f(0.75, 1.1007)$$

$$= 0.5 \left[\frac{(0.75)^2 + (1.1007)^2}{10} \right]$$

$$= 0.0887$$

$$k_4 = hf(x_1 + h, y_1 + k_3)$$

$$= 0.5 f[0.5 + 0.5, 1.0562 + 0.0887]$$

$$= 0.5 f(1, 1.1456)$$

$$= 0.5 \left[\frac{1^2 + (1.1456)^2}{10} \right]$$

$$= 0.1156$$

$$\therefore y_2 = y(1.0) = y_1 + \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4]$$

$$= 1.0569 + \frac{1}{6} [0.0683 + 2 \times 0.0876 + 2 \times 0.0887 + 0.1156]$$

$$= 1.1463$$

Step 3: Putting $n=2$

$$k_1 = hf(x_2, y_2)$$

$$= 0.5 f(1, 1.1463)$$

$$= 0.5 \left[\frac{1 + (1.1463)^2}{10} \right]$$

$$= 0.1157$$

$$k_2 = hf \left(x_2 + \frac{h}{2}, y_2 + \frac{k_1}{2} \right)$$

$$= 0.5 f \left(1 + \frac{0.5}{2}, 1.1463 + \frac{0.1157}{2} \right)$$

$$= 0.5 f [1.25, 1.2042]$$

$$= 0.5 \left[\frac{(1.25)^2 + (1.2042)^2}{10} \right]$$

$$= 0.1506$$

$$k_3 = hf \left(x_2 + \frac{h}{2}, y_2 + \frac{k_2}{2} \right)$$

$$= 0.5 f \left[1 + \frac{0.5}{2}, 1.1463 + \frac{0.1506}{2} \right]$$

$$= 0.5 f (1.25, 1.2216)$$

$$= 0.5 \left[\frac{(1.25)^2 + (1.2216)^2}{10} \right]$$

$$= 0.1527$$

$$k_4 = hf (x_2 + h, y_2 + k_3)$$

$$= 0.5 f [1 + 0.5, 1.1463 + 0.1527]$$

$$= 0.5 f (1.5, 1.299)$$

$$= 0.5 \left[\frac{(1.5)^2 + (1.299)^2}{10} \right]$$

$$= 0.1968$$

$$\begin{aligned}\therefore Y_3 &= Y(1.5) = Y_2 + \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4] \\ &= 1.2463 + \frac{1}{6} [0.1157 + 2 \times 0.1506 + 2 \times 0.1527 \\ &\quad + 0.1968] \\ &= 1.2995\end{aligned}$$

Step 4: Putting $n=3$

$$k_1 = hf(x_3, Y_3)$$

$$= 0.5 \left[\frac{(1.5)^2 + (1.2995)^2}{10} \right]$$

$$= 0.1969$$

$$k_2 = hf\left(x_3 + \frac{h}{2}, Y_3 + \frac{k_1}{2}\right)$$

$$= 0.5 f\left(1.5 + \frac{0.5}{2}, 1.2995 + \frac{0.1969}{2}\right)$$

$$= 0.5 f(1.75, 1.3980)$$

$$= 0.5 \left[\frac{(1.75)^2 + (1.3980)^2}{10} \right]$$

$$= 0.2508$$

$$k_3 = hf \left(x_3 + \frac{h}{2}, y_3 + \frac{k_2}{2} \right)$$

$$= 0.5 f \left[1.5 + \frac{0.5}{2}, 1.2995 + \frac{0.2508}{2} \right]$$

$$= 0.5 f(1.75, 1.4249)$$

$$= 0.5 \left[\frac{(1.75)^2 + (1.4249)^2}{10} \right]$$

$$= 0.2546$$

$$k_4 = hf \left(x_3 + h, y_3 + k_3 \right)$$

$$= 0.5 f \left(1.5 + 0.5, 1.2995 + 0.2546 \right)$$

$$= 0.5 f(2, 1.5541)$$

$$= 0.5 \left[\frac{2^2 + (1.5541)^2}{10} \right]$$

$$= 0.3208$$

$$\therefore y_4 = y(2.0) = y_3 + \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4]$$

$$= 1.2995 + \frac{1}{6} [0.1969 + 2 \times 0.2508 + 2 \times 0.2546 + 0.3208]$$

$$= 1.5543$$

$$\text{So, } y(0) = 1$$

$$y(0.5) = 1.0569$$

$$y(1.0) = 1.1463$$

$$y(1.5) = 1.2995$$

$$y(2.0) = 1.5543$$

51 Solve the system of Linear equations using Gauss Seidal Method:

$$x + 2y + 5z = 20$$

$$5x + 2y + z = 12$$

$$x + 4y + 2z = 15$$

Solution:

Given equations

$$x + 2y + 5z = 20$$

$$5x + 2y + z = 12$$

$$x + 4y + 2z = 15$$

Here, $A = \begin{bmatrix} 1 & 2 & 5 \\ 5 & 2 & 1 \\ 1 & 4 & 2 \end{bmatrix}$, $X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$, $B = \begin{pmatrix} 20 \\ 12 \\ 15 \end{pmatrix}$

Here, $|a_{11}| = 1$, $|a_{22}| = 2$, $|a_{33}| = 2$

We have, $|a_{12}| + |a_{13}| = 2 + 5 = 7 > 1$

$$|a_{21}| + |a_{23}| = 5 + 1 = 6 > 2$$

$$|a_{31}| + |a_{32}| = 1 + 4 = 5 > 2$$

The co-efficient matrix A is not strictly diagonally dominant.

Re-arranging the system we make it strictly diagonally dominant,

$$A = \begin{bmatrix} 5 & 2 & 1 \\ 1 & 4 & 2 \\ 1 & 2 & 5 \end{bmatrix}$$

$$\text{and } B = \begin{pmatrix} 12 \\ 15 \\ 20 \end{pmatrix}$$

The given system (1) becomes

$$5x + 2y + z = 12$$

$$x + 4y + 2z = 15$$

$$x + 2y + 5z = 20$$

From eq (i) we get-

$$x = \frac{1}{5} (12 - 2y - z)$$

$$y = \frac{1}{4} (15 - x - 2z)$$

$$z = \frac{1}{5} (20 - x - 2y)$$

Rewrite eq (iii) as follows-

$$x^{(k)} = \frac{1}{5} (12 - 2y^{(k-1)} - z^{(k-1)})$$

$$y^{(k)} = \frac{1}{4} (15 - x^{(k)} - 2z^{(k-1)})$$

$$z^{(k)} = \frac{1}{5} (20 - x^{(k)} - 2y^{(k)})$$

Taking initial values: $x_0 = 0$, $y_0 = 0$ and $z_0 = 0$

Iteration 1: Putting $k=1$

$$x^{(1)} = \frac{1}{5} (12 - 0 - 0) = 2.4$$

$$y^{(1)} = \frac{1}{4} (15 - 2.4 - 0) = 3.15$$

$$z^{(1)} = \frac{1}{5} (20 - 2.4 - 2 \times 3.15) = 2.26$$

Iteration 2: Putting $k=2$

$$x^{(2)} = \frac{1}{5} (12 - 2 \times 3.15 - 2.26) = 0.688$$

$$y^{(2)} = \frac{1}{4} (15 - 0.688 - 2 \times 2.26) = 2.448$$

$$z^{(2)} = \frac{1}{5} (20 - 0.688 - 2 \times 2.448) = 2.8832$$

Iteration 3: Putting $k=3$

$$x^{(3)} = \frac{1}{5} (12 - 2 \times 2.448 - 2.8832) = 0.8442$$

$$y^{(3)} = \frac{1}{4} (15 - 0.8442 - 2 \times 2.8832) = 2.0974$$

$$z^{(3)} = \frac{1}{5} (20 - 0.8442 - 2 \times 2.0974) = 2.9922$$

Iteration-4: Putting $k=4$

$$x^{(4)} = \frac{1}{5} (12 - 2 \times 2.0974 - 2.9922) = 0.9626$$

$$y^{(4)} = \frac{1}{4} (15 - 0.9626 - 2 \times 2.9922) = 2.0133$$

$$z^{(4)} = \frac{1}{5} (20 - 0.9626 - 2 \times 2.0133) = 3.0022$$

Iteration-5: Putting $k=5$

$$x^{(5)} = \frac{1}{5} (12 - 2 \times 2.0133 - 3.0022) = 0.9942$$

$$y^{(5)} = \frac{1}{4} (15 - 0.9942 - 2 \times 3.0022) = 2.0004$$

$$z^{(5)} = \frac{1}{5} (20 - 0.9942 - 2 \times 2.0004) = 3.001$$

Iteration-6: Putting $k=6$

$$x^{(6)} = \frac{1}{5} (12 - 2 \times 2.0004 - 3.001) = 0.9996$$

$$y^{(6)} = \frac{1}{4} (15 - 0.9996 - 2 \times 3.001) = 1.9996$$

$$z^{(6)} = \frac{1}{5} (20 - 0.9996 - 2 \times 1.9996) = 3.0002$$

Iteration-7: Putting $k=7$

$$x^{(7)} = \frac{1}{5} (12 - 2 \times 1.9996 - 3.0002) = 1.0001$$

$$y^{(7)} = \frac{1}{4} (15 - 1.0001 - 2 \times 3.0002) = 1.9999$$

$$z^{(7)} = \frac{1}{5} (20 - 1.0001 - 2 \times 1.9999) = 3.0000$$

Iteration - 8: Putting $k=8$

$$x^{(8)} = \frac{1}{5} (12 - 2 \times 1.9999 - 3.0000) = 1.0000 \text{ on } 1$$

$$y^{(8)} = \frac{1}{4} (15 - 1.0000 - 2 \times 3.0000) = 2.0000 \text{ on } 2$$

$$z^{(8)} = \frac{1}{5} (20 - 1.0000 - 2 \times 2.0000) = 3.0000 \text{ on } 3$$

After 8th iteration, the difference between iteration 7 and iteration 8 is very negligible.

$\therefore$ The solution of the given system is

$$x=1$$

$$y=2$$

$$z=3 \quad (\text{Ans})$$